

RAS-p110 α signalling in macrophages is required for effective inflammatory response and resolution of inflammation

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Esther Castellano , Alejandro Rosell, Agata Krygowska, Marta Alcón Pérez Alcón, Mathieu-Benoit Voisin, Juan de Paz, Vinothini Rajeeve, Alberto Berral-González, Cristina Cuesta, Otilie Swinyard, Enrique Gabandé-Rodríguez, Julian Downward, Javier De Las Rivas, Pedro Cutillas

Instituto de Biología Molecular y Celular del Cáncer • Queen Mary University of London • Queen Mary University of London-William Harvey Research Institute • Barts Cancer Institute, QMUL • The Francis Crick Institute

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Abstract

Macrophages play a critical role in the body's inflammatory response, and their functions are tightly regulated to ensure optimal immune system function. Here we show that RAS-p110 α signalling, a pathway crucially involved in several biological processes and key for tumorigenesis, also exerts regulatory control over two critical aspects of the inflammatory response: monocyte extravasation during the initial stages of the inflammatory response and lysosomal function during the later stages. Disruption of this pathway in a genetically engineered mouse model, or through pharmacological intervention, leads to impaired inflammatory responses and delays resolution, consequently fostering the emergence of more severe and deleterious acute inflammatory reactions *in vivo*.

Our findings shed light on a previously unknown role of the p110 α isoform in the regulation of immune inflammatory responses in macrophages, providing a better understanding of the molecular mechanisms underlying the complex regulation of macrophage function in this process. Importantly, given recent evidence of the feasibility of activating p110 α with small molecules, our results propose that the RAS-p110 α pathway could serve as a promising pharmacological target for the treatment of chronic inflammation. This potential therapeutic avenue holds great promise for alleviating inflammatory disorders and fostering a better quality of life for patients suffering from such conditions.

eLife assessment

This **useful** study investigates the impact of disrupting the interaction of RAS with the PI3K subunit p110 α in macrophage function in vitro and inflammatory responses in vivo. **Solid** data overall supports a role for RAS-p110 α signalling in regulating macrophage activity and so inflammation, however for many of the readouts presented the magnitude of the phenotype is not particularly pronounced. Further analysis would be required to substantiate the claims that RAS-p110 α signalling plays a key role in macrophage function. Of note, the molecular mechanisms of how exactly p110 α regulating the functions in macrophage have not yet been established.

Introduction

Phosphatidylinositol 3-kinases (PI3K) are a family of lipid kinases that phosphorylate phosphatidylinositides (PtdIns) at the 3'-hydroxyl group¹. Upon activation, PI3K phosphorylates phosphatidylinositol 4,5-bisphosphate (PIP₂) to generate phosphatidylinositol 3,4,5-trisphosphate (PIP₃). PIP₃ serves as a second messenger that recruits proteins containing pleckstrin homology (PH) domains, such as Akt (also known as protein kinase B)^{2,3}. This activation of PI3K regulate various cellular functions, including cell proliferation, growth, survival, motility, inflammation, and metabolism, among others^{1,4}. In macrophages, the activation of PI3K-Akt signalling is crucial to restrict inflammation and to promote anti-inflammatory responses in Toll Like Receptors (TLR)-induced macrophages, contributing to macrophage polarization^{5,6}.

PI3Ks are heterodimeric lipid kinases composed of catalytic and adaptor/regulatory subunits that can be categorized into three classes based on their structures and substrate specificities^{3,7}. Class I catalytic isoforms, including p110 α , p110 β , p110 γ , and p110 δ , play essential roles in integrating signals from growth factors, cytokines, and other environmental cues. While p110 α and p110 β are ubiquitously expressed, p110 δ and p110 γ are largely restricted to the myeloid and lymphoid lineages^{7,9}. While p110 α is primarily associated with cell growth regulation and survival in epithelial cells, it must be considered that this isoform is also expressed in immune cells, including macrophages. The role played by p110 α in macrophages is not well understood, although some studies have suggested that it might regulate the survival and the regulation of the phagocytic activity of macrophages¹⁰. In the context of cancer, impairment of RAS binding to p110 α in macrophages results in reduced recruitment of macrophages to the tumour site^{11,12}. Additionally, this disruption leads to a change in macrophage polarization, favouring a more proinflammatory M1 state¹². These findings suggest that p110 α plays a crucial role in regulating macrophage-dependent functions. However, despite these insights, the precise impact of p110 α on macrophage function and the underlying molecular mechanisms influencing the inflammatory response are not yet fully understood.

Inflammation is a complex and tightly regulated series of events triggered by various stimuli such as pathogens, harmful mechanical and chemical agents, and autoimmune reactions. The inflammatory response primarily occurs in vascularized connective tissues, involving a dynamic interplay of plasma components, circulating cells, blood vessels, and cellular and extracellular factors¹³⁻¹⁷. During inflammation, mediators released by recruited leukocytes orchestrate a response that aims to facilitate tissue repair and protect the body against harmful stimuli¹⁶⁻¹⁸.

Macrophages play a vital role in the inflammatory response by performing functions such as antigen presentation, phagocytosis, and immunomodulation^{16, 17}. Their role begins with the active recruitment of monocytes from the bloodstream to the site of infections¹⁹ where they differentiate into macrophages and recognize microbes and cellular debris through specific mechanisms. Macrophages subsequently actively participate in phagocytosis, a vital process involving the internalization and elimination of pathogens. Microbial destruction predominantly takes place within lysosomes and phagolysosomes^{20–23}. At later stages of inflammation, macrophages contribute to the resolution of inflammation, thus preventing progression from acute to persistent inflammation that would cause additional tissue damage.

In this study we have used a combination of cell biology techniques and animal models to better understand the role of RAS-dependent activation of p110α at the different stages of the inflammatory response. Our findings show that RAS-p110α signalling plays a key role in the initial stages of inflammation, facilitating the extravasation of monocytes from the bloodstream by promoting the necessary cytoskeletal changes. Subsequently, RAS-p110α has a crucial role in lysosomal acidification and activation of cathepsins, which are indispensable for efficient degradation of lysosomal cargo; when these functions are impaired, prolonged acute inflammatory responses and delayed resolution steps are observed. These results significantly enhance our understanding of the complex mechanisms governing the immune response to inflammation, emphasizing the pivotal role played by RAS-p110α signalling in orchestrating proper monocyte extravasation and maintaining optimal lysosomal function.

Results

Disruption of RAS-p110α causes prolonged and more acute responses to inflammatory stress in vivo

Previous data suggested that, in a tumoral setting, somatic disruption of RAS-p110α prevents macrophage recruitment to tumours^{11, 12} and favours polarization of macrophages to a proinflammatory phenotype¹², suggesting a possible role for p110α in macrophage function. To determine whether RAS-dependent activation of p110α participates in innate or adaptive immune responses to inflammation in macrophages, we used bone marrow derived macrophages (BMDMs) from an existing mouse model in which RAS binding to p110α can be disrupted in a tamoxifen inducible manner^{11, 24} (**Supplementary Fig. S1A**). RAS-p110α interaction can be disrupted in this mouse model thanks to the presence of two point mutations in the RAS Binding Domain (RBD) of the *Pik3ca* allele. Additionally, the other *Pik3ca* allele was floxed, allowing tamoxifen activation of a Cre-Recombinase (Cre-ERT2) and inducing the removal of the floxed *Pik3ca* allele, leaving only one *Pik3ca*^{WT} (*Pik3ca*^{WT/-}) or *Pik3ca*^{RBD} (*Pik3ca*^{RBD/-}) allele expressed (**Supplementary Fig. S1B**). *Pik3ca*^{WT/-} and *Pik3ca*^{RBD/-} BMDMs were stimulated towards a proinflammatory state using LPS and IFN-γ and activation of Akt was analysed. Results showed a decrease in the level of Akt activation under inflammatory conditions in BMDMs deficient in RAS-p110α binding, but not in ERK activation, another well-known RAS effector (**Fig. 1A**). Interestingly, the decrease in Akt activation was accompanied by a decrease in the activation of NF-κβ (**Fig. 1B**), a pivotal transcription factor that regulates the expression of pro-inflammatory genes²⁵. These results suggest that RAS-p110α signalling in macrophages may play a crucial role in the initiation, amplification, or resolution of the inflammatory response.

Thus, we next aimed at exploring if disruption of RAS-p110α *in vivo* may cause altered inflammatory responses. For this, we treated 10–12 weeks old *Pik3ca*^{WT/flox} and *Pik3ca*^{RBD/flox} mice with tamoxifen and, two weeks later, performed a paw swelling assay in which zymosan (10 μg/μl) or PBS was injected in the back-hind paw of *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} mice and then paw thickness was measured at regular intervals for 5 days. Paws from the *Pik3ca*^{RBD/-} mice were significantly more inflamed than those from *Pik3ca*^{WT/-} mice from the earlier time points

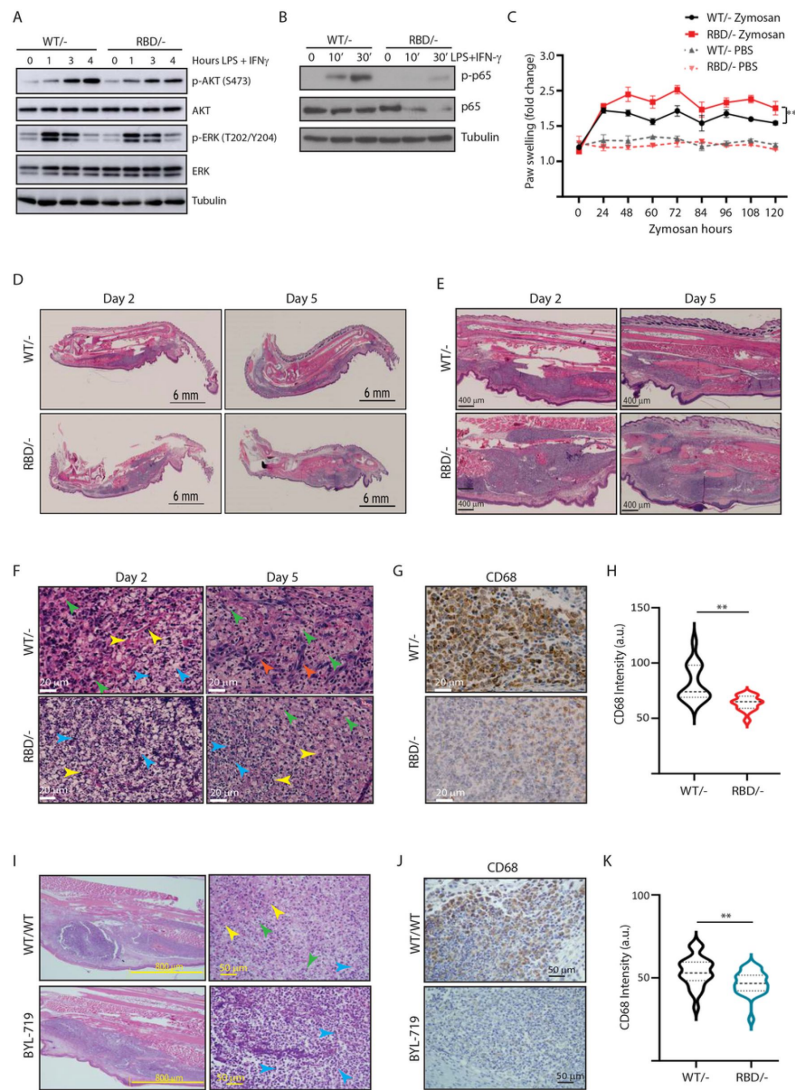


Figure 1

***Pik3ca*^{RBD/-} mice have impaired responses to inflammatory insults.**

(A) Western blotting showing activation of Akt and ERK in *Pik3ca*^{WT/-} and *Pik3ca*^{RBD/-} BMDMs activated towards a pro-inflammatory state with LPS and IFN- γ at the indicated time points; **(B)** Western blotting showing activation of NF- κ B (p65) in *Pik3ca*^{WT/-} and *Pik3ca*^{RBD/-} BMDMs activated towards a pro-inflammatory state with LPS and IFN- γ at the indicated time points; **(C)** *Pik3ca*^{WT/-} and *Pik3ca*^{RBD/-} mice were injected with zymosan or PBS in the back-hind paws and inflammation was measured and plotted over time, *Pik3ca*^{WT/-} PBS n = 6; *Pik3ca*^{RBD/-} PBS n = 5; *Pik3ca*^{WT/-} Zymosan n = 4; *Pik3ca*^{RBD/-} Zymosan n = 4; **(D and E)** Representative H&E images of the inflamed area of *Pik3ca*^{WT/-} and *Pik3ca*^{RBD/-} paws injected with zymosan at the indicated times; **(F)** Representative images of cellularity present in the inflamed abscess of *Pik3ca*^{WT/-} and *Pik3ca*^{RBD/-} paws injected with zymosan at the indicated times. Blue arrows indicate apoptotic polymorphonuclear cells, yellow arrows indicate fibrin deposition, green arrows indicate macrophages and orange arrows indicate activated fibroblasts; **(G)** Representative images of macrophages (CD68 positive cells) present in the inflamed abscess of *Pik3ca*^{WT/-} and *Pik3ca*^{RBD/-} paws injected with zymosan; **(H)** Quantification of macrophages (CD68 positive cells) present in the inflamed abscess of *Pik3ca*^{WT/-} and *Pik3ca*^{RBD/-} paws injected with zymosan; **(I)** Representative H&E images of the inflamed area and cellularity of *Pik3ca*^{WT/WT} paws and *Pik3ca*^{WT/WT} paws treated with BYL-719 injected with zymosan. Blue arrows indicate apoptotic polymorphonuclear cells, yellow arrows indicate fibrin deposition, green arrows indicate macrophages; **(J)** Representative images of macrophages (CD68 positive cells) present in the inflamed abscess of *Pik3ca*^{WT/WT} paws and *Pik3ca*^{WT/WT} paws treated with BYL-719 injected with zymosan; **(K)** Quantification of macrophages (CD68 positive cells) present in the inflamed abscess of *Pik3ca*^{WT/WT} paws and *Pik3ca*^{WT/WT} paws treated with BYL-719 injected with zymosan. Error bars indicate mean $\pm$ SEM. Significance using Student's t test: **p < 0.01.

measured until the end of the experiment (**Fig. 1C**). Furthermore, analysis of blood sedimentation rate, a reliable indicator of systemic inflammation levels²⁶, showed that basal sedimentation ratio (in PBS treated animals) was significantly lower in the *Pik3ca*^{RBD/-} mice (**Supplementary Fig. S1C**); 5 days post zymosan injection, the sedimentation rate increased in animals from both genotypes, but the increase was more pronounced in *Pik3ca*^{RBD/-} mice. These results demonstrate that disruption of RAS-p110 α interaction leads to more acute results in elevated levels of inflammatory mediators in the bloodstream under inflammatory stress.

To further analyse the inflammation reaction caused by zymosan, paws collected at two and five days after zymosan injection and Haematoxylin & Eosin (H&E) studies were carried out. This allowed us to not only investigate the initial inflammatory phase but also track the subsequent resolution phase of inflammation. Paws from mice injected with zymosan for 2 days, presented extended areas of destroyed connective tissue in both *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} paws (**Fig. 1D and E**). However, the inflamed area from the *Pik3ca*^{RBD/-} mice exhibited a larger inflamed area compared to control samples (**Fig. 1D and E**). The inflamed area was significantly reduced after 5 days of zymosan injection, although it remained larger in the paws from *Pik3ca*^{RBD/-} mice (**Fig. 1D and E**).

The cellularity within the inflammatory abscess provides insights into disease severity and progression. Our observations revealed features associated to an acute inflammatory response, such as an inflammatory abscess presenting elevated numbers of polymorphonuclear cells (**Fig. 1F**, blue arrows), mainly neutrophils, and macrophages (**Fig. 1F**, green arrows) with altered cell shape and increased cell death, along with fibrin deposition (**Fig. 1F**, yellow arrows). We observed that *Pik3ca*^{RBD/-} mice exhibited lower numbers of macrophages, a larger necrotic area with increased chromatin remnants and reduced fibrin content than the paws from *Pik3ca*^{WT/-} mice. By day 5, paws from *Pik3ca*^{WT/-} mice showed a significant increase in the number of macrophages (**Fig. 1F**, green arrows), a decrease in polymorphonuclear cells, and a nearly complete resolution of the necrotic area. This indicates the initiation of the resolutive phase. Furthermore, there were high numbers of activated fibroblasts (**Fig. 1F**, orange arrows), suggesting that new connective tissue was being produced. In contrast, paws from *Pik3ca*^{RBD/-} mice exhibited a delayed healing process characterized by a larger central area of polymorphonuclear cells, abundant fibrin deposits, reduced numbers of infiltrating macrophages, and very limited fibroblast activity (**Fig. 1F**). These findings collectively indicate an imbalance in the inflammatory response and a slower progression towards resolution in the absence of RAS-p110 α interaction.

To evaluate the presence of macrophages within the inflammatory lesion, we performed specific immunohistochemical (IHC) analysis using the macrophage-specific marker CD68 in paws from day 5, where more cellular preservation and initiation of the healing process had been observed. A notable decrease in the number of CD68-positive cells was observed in the inflamed abscess region of *Pik3ca*^{RBD/-} mice (**Fig. 1G and H**).

To further confirm the involvement of p110 α signalling in the acute inflammatory response, *Pik3ca*^{WT/WT} mice were subjected to daily treatment with BYL719 (Alpelisib), a specific inhibitor of p110 α isoform. After an initial 48-hour treatment period, the mice received injections of zymosan or PBS into their back-hind paws and were sacrificed 2 days later for analysis. Similar to what we had observed in *Pik3ca*^{RBD/-} mice, the inflamed area in BYL719-treated mice exhibited a larger extension than the inflamed area from non-treated mice (**Fig. 1I** and **Supplementary Fig. S1D**). The central necrotic region was significantly larger in the BYL719-treated mice, and it contained large amount of apoptotic polymorphonuclear cells (**Fig. 1I**, blue arrows) and lower number of macrophages (**Fig. 1I**, green arrows), resembling the observations from *Pik3ca*^{RBD/-} mice. Immunostaining for CD68 revealed a reduction in the number of macrophages present in the inflammatory abscess upon inhibition of p110 α signalling with BYL719 treatment (**Fig. 6J and K**).

In summary, both genetic disruption of RAS - p110 α interaction and pharmacological inhibition of p110 α increase the extension of the inflamed area and the central necrotic region, while reducing macrophage infiltration in a model of zymosan-induced inflammation, demonstrate that the p110 α isoform of PI3K plays a key role in the resolution of inflammatory responses.

Disruption of RAS binding to p110 α impairs the number of monocytes in blood and spleen

Given the decrease in macrophages observed in the inflammatory abscess, we next set out to determine whether disruption of RAS binding to p110 α had an effect on the number of monocytes circulating in the blood of adult mice. *Pik3ca*^{WT/flox} and *Pik3ca*^{RBD/flox} mice were treated with tamoxifen at 12–14 weeks of age and 4 weeks later, blood was collected by cardiac puncture and immune populations were analysed by flow cytometry. We found a decrease in the number of circulating classical (or inflammatory) monocytes (Ly6C^{Hi}/Ly6G⁻/CD11b⁺ cells) (**Fig. 2A**) in *Pik3ca*^{RBD/-} mice and no changes were observed in non-classical (or non-inflammatory) monocytes (Ly6C^{Lo}/Ly6G⁻/CD11b⁺ cells) (**Fig. 2B**). Together with the decrease in inflammatory monocytes, we observed an increase in the number of neutrophils (Ly6C⁻/Ly6G⁺/CD11b⁺ cells) (**Fig. 2C**). We did not detect differences in the numbers of T-cells (CD3⁺, CD8⁺ or CD4⁺) (**Supplementary Fig. S2A** and **S2B**) or B-cells (CD19⁺) (**Supplementary Fig. S2C**) in the blood after the disruption of RAS binding to p110 α .

We next aimed at determining whether splenic immune populations, especially monocytes, would also be altered after disruption of RAS p110 α interaction since splenic monocytes resemble their blood counterparts²⁷. Spleens from *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} mice were collected and immune populations were analysed. As observed in circulating blood, the number of classical monocytes (Ly6C^{Hi}/Ly6G⁻/CD11b⁺) in the spleen decreased after disruption of RAS-p110 α interaction (**Fig. 2D**) and no differences were found in non-classical activated monocytes (Ly6C^{Lo}/Ly6G⁻/CD11b⁺) (**Fig. 2E**). We also checked the levels of resident macrophages present in the spleen and results showed that spleens from the *Pik3ca*^{RBD/-} mice had a decrease in the number of differentiated macrophages (F4/80⁺/CD11b⁺/CD45⁺) (**Fig. 2F**). Analysis of macrophages in the white and red pulp of the spleen indicated that *Pik3ca*^{RBD/-} mice had a significant decrease in the macrophage population in the former region (**Fig. 2G**). No differences were found in the number of granulocytes (**Supplementary Fig. S2D**), B-cells (**Supplementary Fig. S2E**) or T-cells (**Supplementary Fig. S2F**) present in the spleen between *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} mice. There were no differences in spleen size from *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} mice either (**Supplementary Fig. S2G**).

Finally, bone marrow precursors from *Pik3ca*^{RBD/flox} and *Pik3ca*^{WT/flox} mice were differentiated into macrophages in vitro by the addition of macrophage colony stimulation factor (M-CSF) to the culture media for 7 days, and the number of bone marrow-derived macrophages (BMDM) was determined by flow cytometry analysis. No differences were found in the number of BMDMs obtained from *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} mice (**Supplementary Fig. S2H**), suggesting that the disruption of RAS-p110 α signalling do not interfere with the ability of bone marrow precursors to differentiate to macrophages.

Disruption of RAS binding to p110 α impairs monocyte transendothelial extravasation in response to inflammatory cues

The decrease in classical monocytes observed in the inflammatory abscess of paws from *Pik3ca*^{RBD/-} mice may indicate a decreased ability to mount an effective immune response. We carried out transwell assays since they are widely used to quantify transendothelial migration. Fibroblasts were seeded in the lower chamber to provide a continuous supply of Ccl2²⁸, as this cytokine is well known for its ability to drive chemotaxis of myeloid cells under inflammatory conditions²⁹. *Pik3ca*^{RBD/-} or *Pik3ca*^{WT/-} BMDMs were seeded in the upper chamber of the

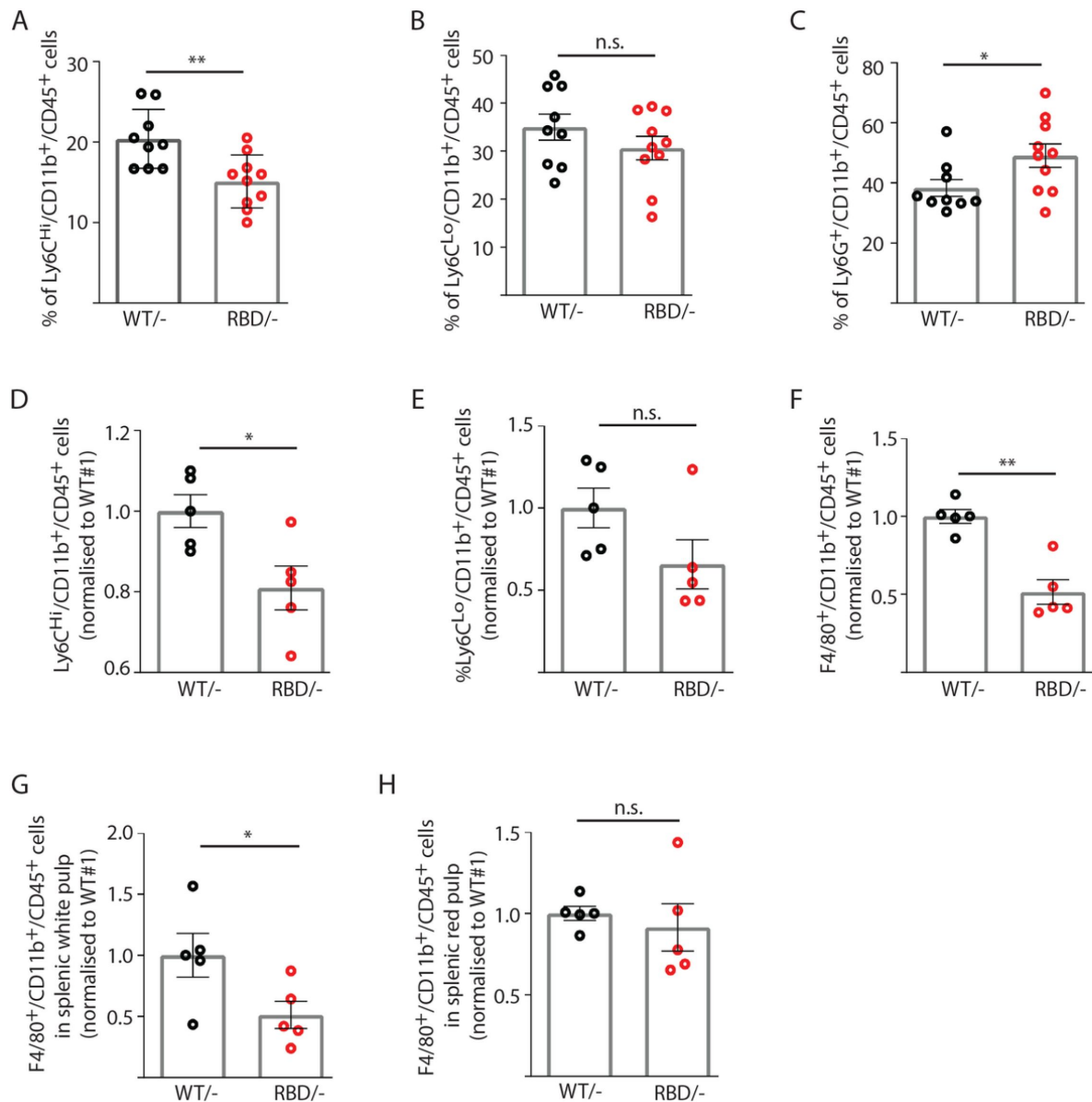


Figure 2

Disruption of RAS-p110 α interaction decreases the number of inflammatory monocytes in blood and spleen.

Twelve-week-old mice were treated with tamoxifen and after four weeks, flow cytometry analysis was performed to determine (A) inflammatory monocytes in circulating blood; (B) alternatively activated monocytes in circulating blood; (C) granulocytes in circulating blood; (D) inflammatory monocytes in spleen; (E) classically activated monocytes in spleen; (F) macrophages in spleen; (G) macrophages in spleen's white pulp; (H) macrophages in spleen's red pulp. Data are presented as percentage of positive cells for the indicated markers. Black dots represent data from *Pik3ca*^{WT/-} mice and red dots represent data from *Pik3ca*^{RBD/-} mice. Each dot represents an individual mouse. Error bars indicate mean ± SEM. Significance using Student's t test: * p < 0.05; **p < 0.01.

transwell. Additionally, we stimulated *Pik3ca*^{RBD/-} or *Pik3ca*^{WT/-} BMDMs with lipopolysaccharide (LPS) and Interferon gamma (IFN-γ) to mimic/recapitulate the proinflammatory phenotype typically associated with bacterial infection that causes macrophage activation³⁰. Results showed that in both unstimulated and LPS + IFN-γ stimulated BMDMs, a lower number of *Pik3ca*^{RBD/-} BMDMs were able to go through the trans-well pore membranes compared to *Pik3ca*^{WT/-} BMDMs (**Fig. 3A**). As expected, when BMDMs were stimulated towards a proinflammatory phenotype, a decrease in their migratory ability is observed³¹.

Extravasation entails the migration of monocytes through the endothelium^{32, 33}, so we conducted random migration assays of *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} BMDMs growing in matrigel coated plates under unstimulated conditions or activated towards an inflammatory phenotype by addition of LPS and IFN-γ to the media. Analysis of the data revealed that, under pro-inflammatory conditions, *Pik3ca*^{RBD/-} BMDMs displayed a decrease in migration speed (**Fig. 3B**). Migration was also evaluated in *Pik3ca*^{WT/WT} BMDMs treated with BYL-719. Results confirmed a decrease in migration speed of macrophages upon treatment with BYL-719 (**Fig. 3B**) indicating that inhibition of p110α, either genetically or pharmacologically, reduces the ability of macrophages to migrate under inflammatory conditions.

To determine if disruption of RAS binding to p110α impairs monocyte ability to extravasate through the endothelium in vivo in response to an inflammatory stress, we analysed monocyte extravasation through the mesenteric vein in response to intraperitoneal Ccl2 injection. It is well established that loss of p110α function leads to significant impairment in endothelial and lymphatic system^{12, 34}, so in order to determine if monocytes from *Pik3ca*^{RBD/-} mice presented an alteration in extravasation, we generated chimeric mice in which only the bone marrow were defective in RAS binding to p110α (**Fig. 3C**). For this, bone marrow from *Pik3ca*^{RBD/-} or *Pik3ca*^{WT/-} mice was injected through the tail vein of irradiated LysM-GFP donor mice³⁵. Engraftment of *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} bone marrow could be followed by disappearance of the eGFP signal in donor mice (**Fig. 3D** and **Supplementary Fig. S3A**). Neutrophils (but no monocytes) were depleted using an anti-GR1 antibody (**Fig. 3E** and **Supplementary Fig. S3B**) to avoid interference with monocyte extravasation. Intraperitoneal injection of Ccl2 was performed in the peritoneum of the chimera mice to induce extravasation of monocytes through the mesenteric vein and rolling, adhesion and extravasation was measured by intravital microscopy. Flow cytometry analysis demonstrated no differences in Ccr2 expression between *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} monocytes (**Supplementary Fig. S3C**). We observed that blocking RAS-p110α interaction did not induce a decrease in the number of rolling monocytes (**Fig. 3F**) or in the number of monocytes that adhere to the endothelium (**Fig. 3G**). However, when extravasation was measured, data showed that disruption of RAS binding to p110α caused a significant decrease in the number of monocytes that were capable of extravasating through the endothelium of the mesenteric vein (**Fig. 3H**).

Disruption of RAS-p110α activation in macrophages induces changes in cytoskeleton reorganization

During transendothelial migration, leukocytes undergo cytoskeletal rearrangements that allow them to squeeze through the tight spaces between endothelial cells and enter the underlying tissue^{36, 37}. Therefore, we next sought to determine whether the differences observed in *Pik3ca*^{RBD/-} BMDMs extravasation and migration were attributable to altered cytoskeletal dynamics. To do so, we first aimed at examining cell shape and spread in *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} BMDMs in pro-inflammatory conditions after LPS + IFN-γ treatment. Treatment with LPS + IFN-γ induced an increase in cell spread in both *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} BMDMs when compared to their respective unstimulated counterparts (**Fig. 4A and 4B**). However, cell spread in LPS + IFN-γ activated *Pik3ca*^{RBD/-} BMDMs was significantly decreased compared to that observed *Pik3ca*^{WT/-} BMDMs (**Fig. 4A and 4B**). Additionally, *Pik3ca*^{RBD/-} BMDMs were more elongated and did not acquire the typical rounded shape known to be induced in macrophages

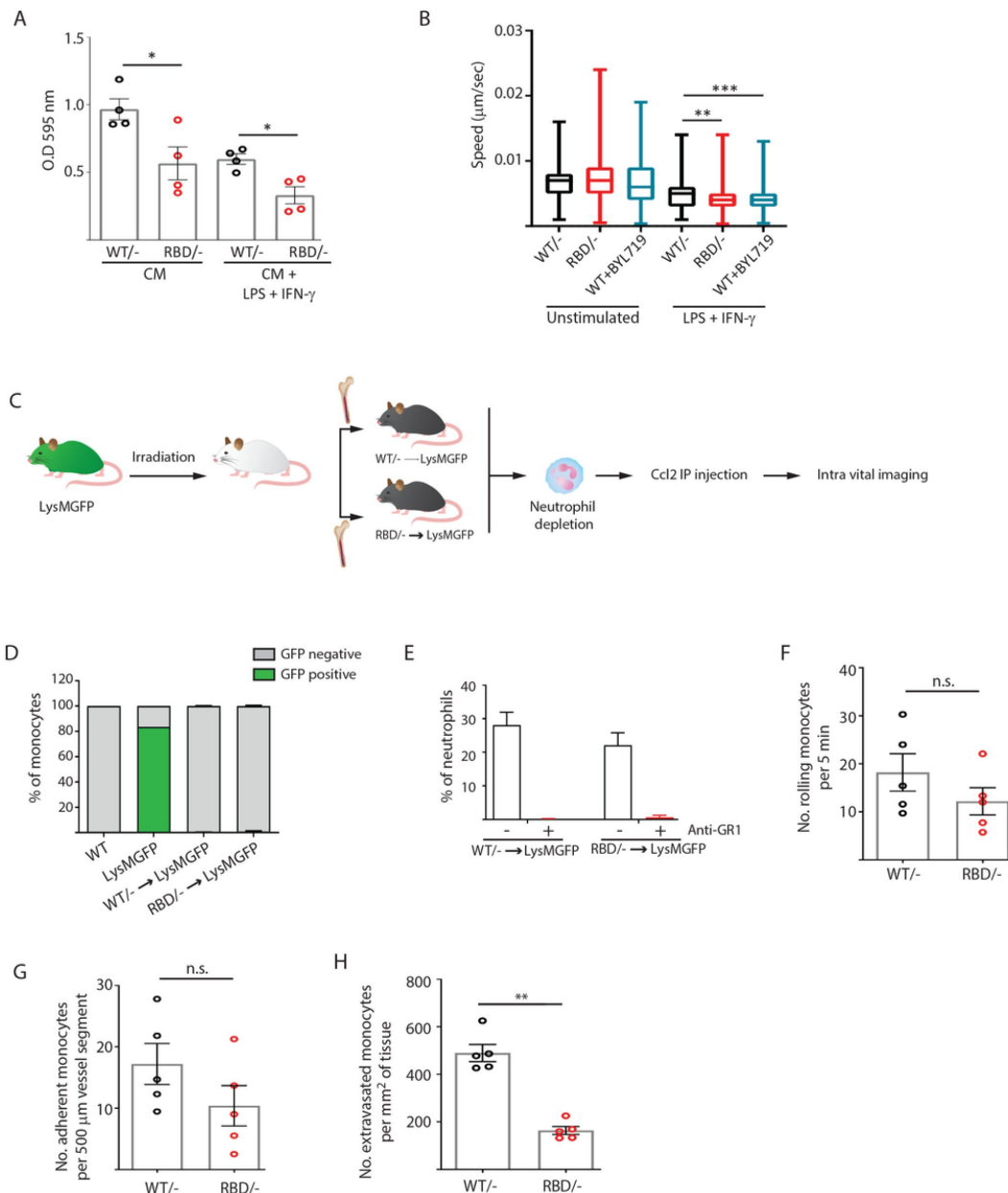


Figure 3

Disruption of RAS-p110 α interaction impairs transendothelial extravasation to sites of inflammation.

(A) Graph indicating the quantification of *Pik3ca*^{RBD/-} and *Pik3ca*^{RBD/-} BMDMs passing through 8 μ m membrane pore transwells for 24 hours. Membrane attached macrophages on the lower part of the transwell were stained with crystal violet and quantified. Each dot represents an independent experiment. Error bars indicate mean $\pm$ SEM; (B) Random migration of *Pik3ca*^{WT/-}, *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/WT} BMDMs treated with BYL719 (500 ng/ml) was analysed by time-lapse video microscopy and cell tracing in the presence or absence of LPS (100 ng/ml) and IFN- γ (20 ng/ml); (C) Schematic representation of myeloid chimera generation strategy; (D) Graph showing level of bone marrow reconstitution with *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} myeloid lineage; (E) Graph quantifying the number of neutrophils in the blood of chimera mice treated with anti-GR1 (25 μ g/mouse/day); (F-H) Intra vital imaging quantification of the number of: (F) rolling monocytes per 5 minutes; (G) adherent monocytes per 500 μ m of vessel segment; (H) extravasated monocytes per mm² of tissue. Each dot represents an individual mouse.

Statistical significance was obtained using Mann-Whitney test: n.s. non-significant; *p < 0.05; **p < 0.01; ***p < 0.001; ****p < 0.0001.

after treatment with LPS + IFN- γ ³⁸ (Fig. 4A and 4C). We also analysed cell height and results showed that *Pik3ca*^{RBD/-} BMDMs are higher both in unstimulated conditions and after LPS + IFN- γ stimulation (Fig. 3D).

Actin is a bona fide regulator of cell shape³⁹, so we next aimed at exploring the actin cytoskeleton in *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} BMDMs. Actin fractionation assays were carried out in unstimulated or LPS + IFN γ activated *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} BMDMs. Our results revealed that disruption of RAS-p110 α binding caused a decrease in the F-actin pool in unstimulated BMDMs (Fig. 4E), with no changes observed in the G-actin pool when compared to matched controls (Fig. 4F). Analysis of actin dynamics after activation with LPS + IFN γ showed that, in proinflammatory conditions, the F-actin pool is increased, as expected⁴⁰ (Fig. 4E). However, actin polymerization was significantly active in *Pik3ca*^{RBD/-} BMDMs, leading to a striking increase in F-actin when compared to that observed in controls (Fig. 4E). In parallel, a decrease in the G-actin pool in *Pik3ca*^{RBD/-} BMDMs was observed, suggesting an increase in the stabilization of F-actin after disruption of RAS binding to p110 α (Fig. 4F).

We performed Western blot analysis to assess the expression and activation of proteins involved in actin polymerization dynamics. Interestingly, we found elevated levels of serine 3 phosphorylated Cofilin in *Pik3ca*^{RBD/-} BMDMs both in unstimulated and in proinflammatory conditions (Fig. 4G). Phosphorylation of Cofilin at Ser3 hampers its ability to depolymerize F-actin, resulting in the stabilization of actin and affecting actin filament dynamics. The increase in phosphorylated Cofilin, thus, would explain the increase in actin stabilization observed in *Pik3ca*^{RBD/-} BMDMs. Additionally, we also observed an upregulation in the phosphorylation of LIM domain kinase (LIMK), a known kinase responsible for phosphorylating Cofilin at Ser3 (Fig. 4G), suggesting that the heightened Ser3 phosphorylation of Cofilin may be LIMK-mediated.

Overall, our findings provide strong evidence indicating that the impaired extravasation of monocytes can be primarily attributed to disruptions in cytoskeletal dynamics.

Disruption of RAS-p110 α activation impacts the secretome of macrophages

We next wondered whether the absence of RAS binding to p110 α affects the ability of BMDMs to acquire a pro-inflammatory state characterized by increased expression of markers such as CD80, CD86, and MHCII^{41, 42}. To assess this, *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} BMDMs were stimulated with LPS + IFN- γ , followed by flow cytometry analysis. Expression levels of CD80 (Supplementary Fig. S4A), CD86 (Supplementary Fig. S4B), and MCHII (Supplementary Fig. S4C) were examined. No significant differences were observed in the expression levels of any of these markers between the two genotypes under study.

During the inflammatory response, macrophages release a diverse array of bioactive molecules that have a profound impact on immune regulation, tissue repair, and disease progression. Thus, we analysed the secretome of *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} BMDMs both under steady state condition and during phagocytosis. We utilized apoptotic LKR10 cells, a murine lung cancer cell line, as the substrate in our phagocytosis assay. LKR10 cells were exposed to cisplatin for 16 h and apoptosis was confirmed by Western Blot (Supplementary Fig. S4D). Our goal was to create a more physiologically relevant experimental setting that more closely mimics the complex nature of the inflammatory response. For the secretome analysis, macrophages were incubated with or without apoptotic cells for 16 h. Culture supernatants were collected, clarified, and subjected to label-free quantitative proteomics analysis.

A total of 127 peptides corresponding to 105 proteins (Supplementary Table 1) showed differential expression between *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} BMDM secretomes at steady state conditions. Additionally, 359 peptides corresponding to 210 proteins (Supplementary Table 2)

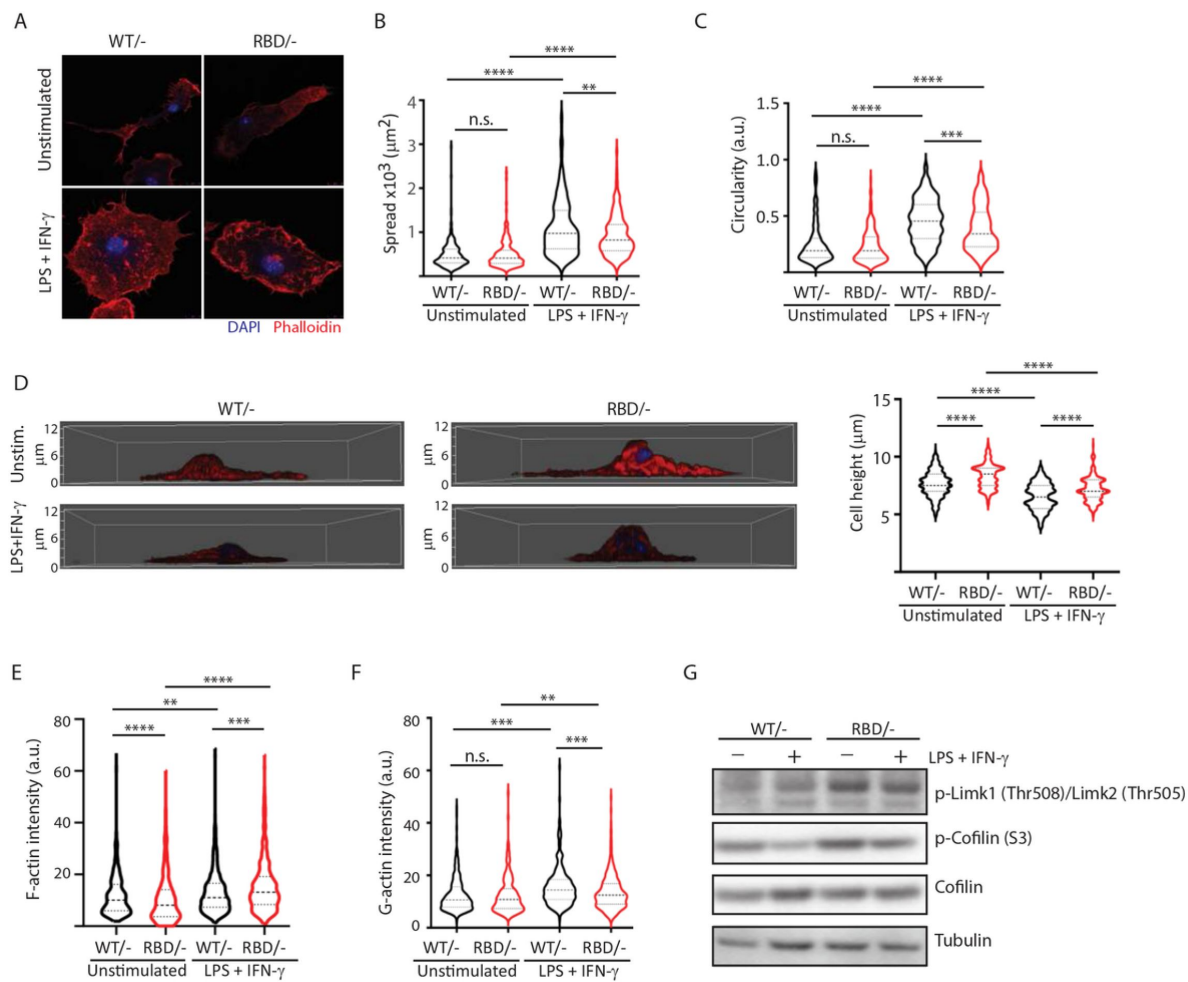


Figure 4

Disruption of RAS-p110a signalling impairs BMDM ability to remodel their cytoskeleton in response to inflammatory cues.

(A) Representative images of *Pik3ca*^{WT/-} and *Pik3ca*^{RBD/-} BMDMs unstimulated or activated with LPS+ IFN-γ; (B) violin plot quantifying spread area. IF images of *Pik3ca*^{WT/-} and *Pik3ca*^{RBD/-} BMDMs co-stained with phalloidin and DAPI were used to analyse spread area. Results are represented as a violin plot. Three independent biological replicates were analysed (n ≥ 250 total cells); (C) Quantification of cell circularity of *Pik3ca*^{WT/-} and *Pik3ca*^{RBD/-} BMDMs unstimulated or activated with LPS+ IFN-γ. Quantification was performed using the same images from panel (A); (D) Representative 3D projections of *Pik3ca*^{WT/-} and *Pik3ca*^{RBD/-} BMDMs unstimulated or activated with LPS+IFN-γ and violin plot showing quantification of the corresponding cell height. 3D projections and cell height were analysed using same images used in panel (A). (E) Violin plot representing F-actin pool in *Pik3ca*^{WT/-} and *Pik3ca*^{RBD/-} BMDMs unstimulated or activated with LPS+ IFN-γ. Three independent biological replicates were analysed (n ≥ 250 total cells); (F) Violin plot representing G-actin pool (measured by DNase staining) in *Pik3ca*^{WT/-} and *Pik3ca*^{RBD/-} BMDMs unstimulated or activated with LPS+ IFN-γ. Three independent biological replicates were analysed (n ≥ 250 total cells); (G) *Pik3ca*^{WT/-} and *Pik3ca*^{RBD/-} BMDMs were activated with LPS+ IFN-γ and western blots were performed to determine expression levels of the indicated proteins.

Statistical significance was obtained using Mann-Whitney test: n.s. non-significant; *p < 0.05; **p < 0.01; ***p < 0.001; ****p < 0.0001.

were present a significantly different levels in the secretomes of *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} BMDM during phagocytosis of apoptotic cells. Next, we compared these peptides with the list of secreted proteins available at The Human Protein Atlas, and removed those that do not correspond to secreted proteins. After this step, 18 proteins were found to be differentially secreted by *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} BMDM in steady state conditions (Fig. 5A), and 38 by *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} BMDM during phagocytosis (Fig. 5B and Supplementary Fig. S4E and S4F). Surprisingly, most proteins were secreted at lower levels in *Pik3ca*^{RBD/-} BMDMs, independently of the conditions under study. 12 proteins were differentially secreted in both experimental conditions under study (Fig. 5C).

Differentially secreted proteins are involved in two main biological processes: complement activation (C1qa, C1qb and C1qc connected through physical interaction and C3, C9 and Cfb through coexpression) and lysosome function, mainly cathepsins (Ctsd, Ctsb, Ctsz, Cst3, Psap, Anxa1 and Gsn). Together, the complement cascade and lysosome function work in concert to provide an effective defence against pathogens and promote overall maintenance of cellular homeostasis^{22, 43-45} and data from the secretome analysis of *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} BMDMs suggests that RAS activation of p110α may play a crucial role in the regulation of both response pathways.

Disruption of RAS-p110α signalling leads to altered lysosomal function

We next aimed to functionally validate the proteomics data suggesting that RAS-p110α activation regulates lysosomal function in macrophages. First, we performed immunofluorescence analysis of lysosomal-associated membrane protein 1 (LAMP1) to assess lysosomal biogenesis and function in *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} BMDMs. Analysis of LAMP1 in *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} BMDMs showed a decrease in LAMP1 expression in steady state conditions (Fig. 6A) suggesting a decrease in the number of lysosomes after disruption of RAS-p110α interaction. However, *Pik3ca*^{RBD/-} showed increased levels of Lamp1 expression when subjected to phagocytosis of apoptotic cells, suggesting an increase in the number of phagolysosomes present in these cells.

We hypothesised that the increase in Lamp1 expression in BMDMs lacking RAS-p110α interaction during phagocytosis could be attributed to aberrant lysosomal function and phagolysosome retention. Thus, we next evaluated lysosomal activity by using the lysosomotropic dye lysotracker red coupled with epifluorescence analysis, since it is specifically taken up by acidic organelles. As such, its accumulation is proportional to the number of acidic vesicles. Data analysis showed that disruption of RAS-PI3K in BMDMs leads to a significant decrease in lysotracker uptake, both in unstimulated conditions and also after activation with LPS + IFN-γ (Fig. 6B). Similar results were obtained with control BMDMs treated with BYL719, the p110α specific inhibitor. Data analysis showed a decrease in the uptake of lysotracker after p110α inhibition (Fig. 6B), further suggesting that loss of p110α function in BMDMs results in altered lysosomal pH. To confirm that lysosomal pH of *Pik3ca*^{RBD/-} BMDMs was less acidic, we next stained *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} BMDMs with Green lysosensor, a pH-sensitive dye that exhibit a pH-dependent increase in fluorescent intensity upon lysosomal acidification. As shown in Fig. 6C, *Pik3ca*^{RBD/-} BMDMs presented attenuated fluorescent intensity both in unstimulated and activated when compared with that in the control group, indicating that their lysosomal content is less acidic than in *Pik3ca*^{WT/-} BMDMs.

Considering that *Pik3ca*^{RBD/-} lysosomes were less acidic, we next investigated the expression level and activation of some of the cathepsins identified in the secretome analysis (Fig. 5) by western blotting. Cathepsins play a critical role in lysosomal protein degradation. They are initially synthesized as inactive precursors and are activated through proteolysis in the lysosome at a low pH⁴⁶. We found a reduction in the expression and activation of cathepsin D and cathepsin B in

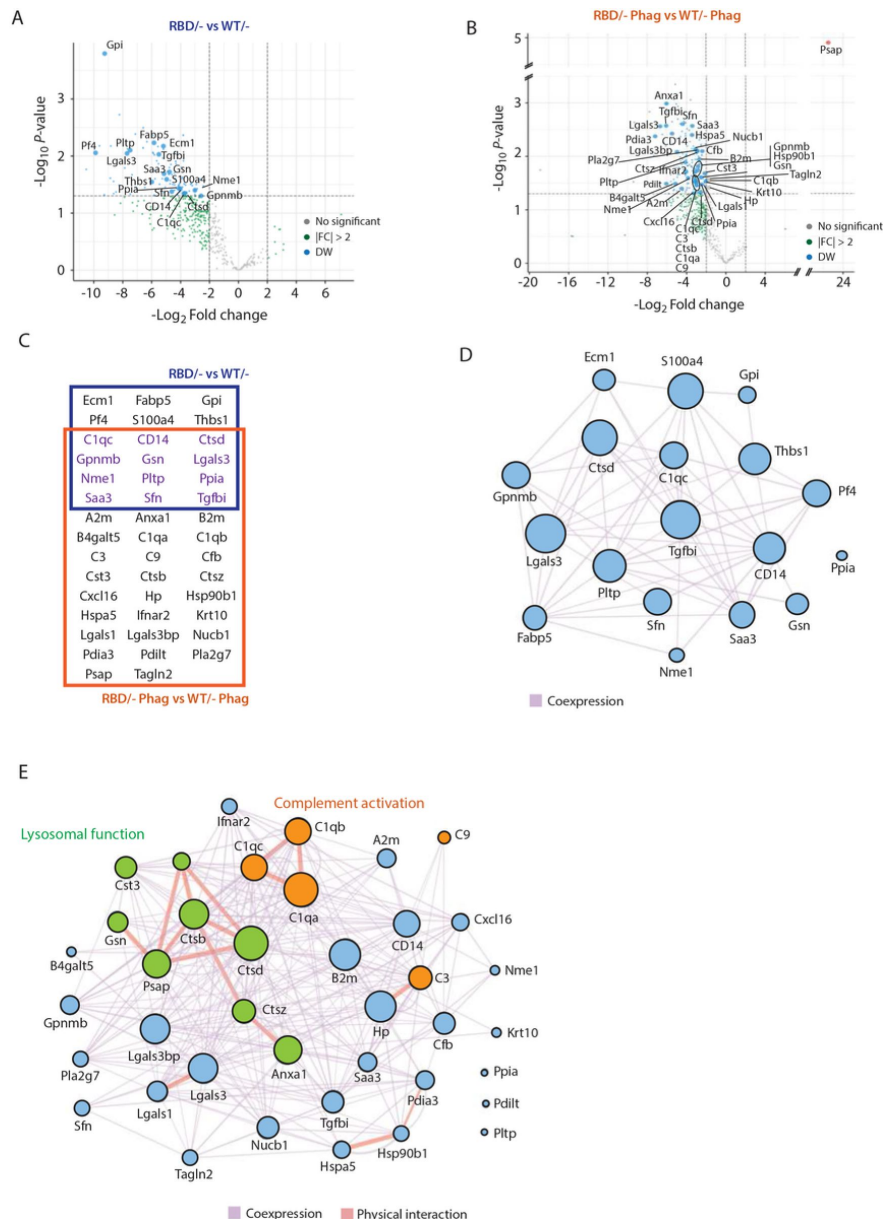


Figure 5

Secretome analysis of *Pik3ca*^{RBD/-} BMDMs suggested a defect in complement activation and lysosomal function.

(**A** and **B**) Volcano plots of secretome analysis from *Pik3ca*^{WT/-} and *Pik3ca*^{RBD/-} BMDMs in non-stimulated conditions (**A**) or phagocytosing apoptotic cells (**B**). The x-axis shows the log₂ FC of each identified protein and the y-axis the corresponding -log₁₀ P value. Statistically significant peptides with FC ≥ 2 and p-value < 0.05 are in blue; peptides that do not pass this threshold are in grey; peptides with FC ≥ 2 but p-value ≥ 0.05 are in green; (**C**) Venn diagram showing the overlap of *Pik3ca*^{RBD/-} vs *Pik3ca*^{WT/-} BMDMs differentially expressed proteins at rest and during phagocytosis of apoptotic cells. Proteins displayed in the blue square represents peptides differentially expressed in resting BMDMs, the brown square represents peptides differentially expressed during phagocytosis, and the peptides differentially expressed in both conditions are displayed in purple; (**D** and **E**) Network analysis of significantly expressed proteins identified in the secretome analysis of *Pik3ca*^{RBD/-} vs *Pik3ca*^{WT/-} BMDMs in steady state conditions (**D**) or during phagocytosis of apoptotic cells (**E**). The nodes represent individual proteins, and the edges represent known interactions between proteins, either co-expression (purple) or physical interaction (thick pink). The size of each node reflects the significance of differential expression. Proteins in green have been implicated in lysosomal function, while proteins in orange are members of the complement cascade.

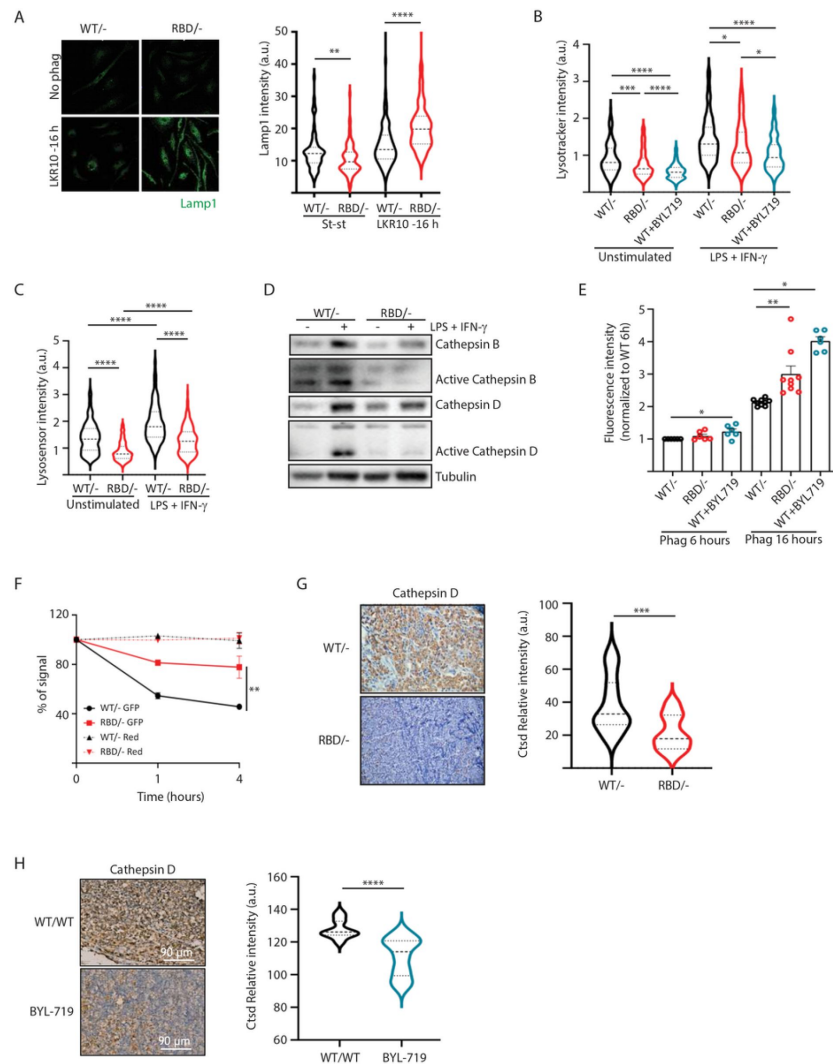


Figure 6

Disruption of RAS-p110a activation in BMDMs leads to abnormal lysosomal function

Statistical significance was obtained using Mann-Whitney test: *p < 0.05; **p < 0.01; ***p < 0.001; ****p < 0.0001.

(A) Representative IF images and quantification analysis of lamp intensity in *Pik3ca*^{WT/-} and *Pik3ca*^{RBD/-} BMDMs in steady state conditions and during phagocytosis of apoptotic LKR10 cells. Three independent biological replicates were analysed (n ≥ 250 total cells). Error bars indicate mean ± SEM; **(B)** Quantification analysis of lysosomal function in *Pik3ca*^{WT/-}, *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/WT} BMDMs treated with BYL719 BMDMs unstimulated or activated with LPS + IFN-γ using LysoTracker staining. Three independent biological replicates were analysed (n ≥ 250 total cells); **(C)** Violin plot displaying lysosome pH acidity in unstimulated and LPS+IFN-γ stimulated *Pik3ca*^{WT/-} (black), *Pik3ca*^{RBD/-} (red) and *Pik3ca*^{WT/WT} BMDMs treated with BYL719 (Blue) BMDMs, as determined by LysoSensor staining. Three independent biological replicates were analysed (n ≥ 250 total cells); **(D)** *Pik3ca*^{WT/-} and *Pik3ca*^{RBD/-} BMDMs were activated with LPS (100 ng/ml) and IFN-γ (20 ng/ml) and expression and activation of Cathepsin B and D were blotted; **(E)** Quantification of phagocytosed apoptotic cells in *Pik3ca*^{WT/-}, *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/WT} BMDMs treated with BYL719 BMDMs. Apoptotic cells were labelled with a red tracker and allowed to be phagocytosed by control, *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/WT} BMDMs treated with BYL719 for 16 hours, measuring the amount of cell tracker that was internalized at different time points. Each dot of the graph indicates a different independent experiment. Error bars indicate mean ± SEM; **(F)** Graph displaying the ability of *Pik3ca*^{WT/-} and *Pik3ca*^{RBD/-} BMDMs to degrade GFP-labelled (pH sensitive) and red tracker-labelled (pH insensitive) apoptotic cells. Three independent biological replicates were analysed. Error bars indicate mean ± SEM; **(G)** Representative image and quantification of Cathepsin D expression in the inflammatory abscess of *Pik3ca*^{WT/-} and *Pik3ca*^{RBD/-} paws injected with zymosan; **(H)** Representative image and quantification of Cathepsin D expression in the inflammatory abscess of paws from control and *Pik3ca*^{WT/WT} mice treated with BYL719 injected with zymosan.

Pik3ca^{RBD/-} BMDMs upon stimulation with LPS + IFN-γ (Fig. 6D). This observation suggests that the impaired lysosomal pH observed in *Pik3ca*^{RBD/-} BMDMs could potentially account for the observed decrease in cathepsin activity.

Acidification of lysosomes and cathepsin activation are critical steps for activation of the resolute stage of the inflammatory response, so we next evaluated the ability of *Pik3ca*^{RBD/-} BMDM's lysosomal compartment to degrade internalised particles. We set up a phagocytosis assay in which *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} BMDMs were labelled with a green tracker and were allowed to engulf apoptotic cells that were labelled with Cell tracker Red CMTPIX. We expected that, if disruption of RAS-p110α interaction impairs lysosomal degradation of engulfed material, after a certain period of time *Pik3ca*^{RBD/-} BMDMs presented an accumulation of the red tracker from non-digested apoptotic cells. Flow cytometry analysis showed that *Pik3ca*^{RBD/-} BMDMs accumulate significantly higher levels of red tracker signal from apoptotic cells than *Pik3ca*^{WT/-} BMDMs over time, suggesting that disruption of RAS-p110α in BMDMs causes an accumulation of non-digested apoptotic cells (Fig. 6E). Furthermore, inhibiting p110α with BYL719 also resulted in a significant accumulation of apoptotic cell debris within BMDMs over time, confirming the involvement of p110α signalling in the digestion of lysosomal content.

To further confirm the impairment of phagocytosis in *Pik3ca*^{RBD/-} BMDMs to degrade phagocytosed particles, we set up another phagocytosis assay in which *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} BMDMs would phagocytose apoptotic cells that were transduced with GFP (which is pH sensitive⁴⁷ [↗](#)) and also labelled with Celltracker Red CMTPIX (pH insensitive). *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} BMDMs were allowed to engulf these apoptotic cells for 16 h and after this time, apoptotic cells were eliminated by washes and BMDMs were analysed by flow cytometry at different time points to measure GFP signal. Our results showed that GFP signal is lost significantly faster in control BMDMs than in *Pik3ca*^{RBD/-} BMDMs (Fig. 6F). As expected, we did not observe any decay in the red tracker signal. Collectively, these data evidence that *Pik3ca*^{RBD/-} BMDMs exhibit a significant impairment in removing engulfed particles, that can be attributed to the absence of lysosome acidification.

We have shown a delayed clearance of apoptotic cells in *Pik3ca*^{RBD/-} mice after zymosan injection (Fig. 1). This, together with previous data took us to analyse the levels of Cathepsin D in the inflamed abscess from the paws of *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} mice. Results showed a significant decrease in the levels of Cathepsin D in the inflamed area of *Pik3ca*^{RBD/-} mice (Fig. 6G) and BYL719-treated mice (Fig. 6H).

In summary, our findings underscore the crucial role of RAS-p110α signalling axis in maintaining the balance of the inflammatory response and promoting timely resolution, providing further evidence for the significant involvement of RAS-p110α signalling in the response to inflammatory stimuli.

Discussion

In this study, we provide compelling evidence that disruption of RAS-p110α signalling or chemical inhibition of p110α impairs response to inflammatory stresses due to defects in both monocyte extravasation during the early stages of the inflammatory response and decreased lysosomal function during the later stages.

Monocyte extravasation during the inflammatory response is a critical step that allows immune cells to reach the site of infection or injury¹⁹ [↗](#). Impairment on this process causes slower or inadequate immune response, as macrophages are crucial for detecting and engulfing pathogens or cellular debris at the site of inflammation, as well as delayed resolution of inflammation, resulting in prolonged inflammation and potential tissue damage⁴⁸ [↗](#), ⁴⁹ [↗](#). Our data shows that

RAS binding to p110 α is involved in macrophage extravasation by modulating actin dynamics through LIMK-cofilin signalling, two well-known regulators of actin polymerization that facilitate the disassembly of actin filaments⁵⁰. During monocyte extravasation, actin filaments form actin-rich protrusions that are essential for monocyte migration across the endothelium^{37, 51}. Our data show that, during extravasation, RAS-p110 α signalling regulates actin depolymerization so monocytes are able to squeeze through endothelial cells^{37, 39}.

Although future research should aim to elucidate the specific mechanisms by which RAS-p110 α signalling regulates this process, it is plausible to consider that RAS-p110 α regulation of cofilin is achieved through members of the Rho-GTPases. The relationship between RAS-PI3K signalling and Rho-GTPases is well established⁵², with several interconnection points between both signalling pathways. LIMK phosphorylation can occur through downstream effectors of phosphoinositide 3-kinase (PI3K), such as phosphorylated p21-activated kinase (PAK)⁵³ or Rho kinase (ROCK)⁵⁴, which are regulated by Rho GTPases. Furthermore it has been described that PI3K inhibition causes an increase in LIMK and cofilin phosphorylation⁵⁵. A better understanding of the specific molecular mechanisms connecting LIMK phosphorylation and RAS-p110 α , will offer a comprehensive view of cell motility regulation. This knowledge may have broad implications for therapeutic development, disease management, tissue engineering, since it represents a promising avenue for targeted interventions and more personalized approaches in medicine.

Given the role of RAS-p110 α interaction in cell extravasation, the decrease in the number of classical monocytes in the *Pik3ca*^{RBD/-} mice may be due to a defect in the extravasation process. Monocyte extravasation shares many features with monocyte egression from the bone marrow. During monocyte egress, monocytes also undergo changes in cytoskeleton dynamics to detach from the sinusoidal endothelial cells and to migrate through the endothelial fenestrae and basement membrane into the bone marrow sinusoids⁵⁶. This result may also explain, at least partially, the lack of macrophage recruitment to lung tumours found in previous studies using the RBD mouse model^{11, 12}.

Our findings also revealed a crucial role of RAS-p110 α activation in the acute phase of the inflammatory response by regulating effective lysosomal degradation of phagocytosed and engulfed material. Phagocytosis constitutes a vital step in the inflammatory response, whereby phagosomes bind to lysosomes to form phagolysosomes in which pathogens are eradicated to facilitate an appropriate host response. This response encompasses antigen presentation to engage T-cell responses, secretion of inflammatory mediators that guide the adaptive immune response, and initiation of tissue repair mechanisms^{22, 45}. Our data provide evidence that the activation of RAS-p110 α signalling pathway is involved in the critical process of lysosomal acidification, which is essential for the efficient degradation of internalized particles and the activation of proteolytic enzymes, ultimately resulting in the formation of fully functional lysosomes. Consequently, lack of lysosome acidification impairs the expression and activation of important proteases such as cathepsin B and cathepsin D. The proper functioning of lysosomes is essential for mounting a robust response to inflammatory stress. Lysosomes play a central role in the breakdown of pathogens and dead cells by providing the necessary degradative enzymes and maintaining an acidic environment that facilitates the degradation of engulfed particles^{20, 21}. When lysosomal function is compromised in macrophages, the degradation of phagocytosed material becomes impaired, leading to the accumulation of toxic debris. This accumulation subsequently triggers inflammation and causes damage to the surrounding tissues, leading to chronic inflammation and appearance of lysosomal storage diseases (LSD)^{57, 58}.

The possible functional link between the increased actin polymerization and lysosomal dysfunction observed in *Pik3ca*^{RBD/-} mice remains an unanswered question. Lysosome acidification and actin polymerization are tightly interconnected processes that have pivotal roles in numerous cellular functions⁵⁹⁻⁶¹. Studies have demonstrated that lysosome acidification can influence actin polymerization dynamics through the activation of specific actin-binding

proteins and the modulation of actin-regulatory proteins⁵⁹,⁶². Conversely, disruption in lysosome acidification, such as impaired proton pump activity or lysosomal storage disorders, have been associated with changes in actin polymerization and the organization of the cytoskeleton. Notably, it has been reported that Cofilin plays a crucial role in assembling the macromolecular components of the acidification machinery in nascent endosomes by depolymerizing F-actin⁶³. Therefore, the intricate relationship between lysosome acidification and actin polymerization suggests a potential reciprocal influence, where perturbations in one process could impact the other, highlighting the need for further investigation to unravel the precise interplay between these two essential cellular mechanisms. Further investigation is required to determine the precise regulatory mechanisms by which p110α influences both lysosome acidification and actin polymerization, whether it occurs in a linear manner or through separate pathways.

In summary, our findings offer valuable insights into the underlying mechanisms governing the immune response to inflammation, emphasizing the critical role of RAS-p110α signalling in the context of inflammation. Given the relevance of p110α signalling in the resolution of inflammation and the persistent dysregulated inflammation seen in diseases such as rheumatoid arthritis, inflammatory bowel disease, psoriasis, and systemic lupus erythematosus, among others, exploring the potential of p110α as a therapeutic target in those indications is intriguing. The identification of a p110α small molecule activator⁶⁴ provides a tool to evaluate the therapeutic potential of transiently activating p110α in these diseases by testing if enhancing p110α function promotes the resolution of inflammation and potentially alleviate the symptoms and progression of these inflammatory conditions.

Methods

Animal studies

For removal of *Pik3ca*-floxed allele, *Pik3ca*^{RBD/lox} and *Pik3ca*^{WT/flox} mice⁶⁴ were given 3.2 mg tamoxifen (Sigma) dissolved in 80 μl of corn oil by oral gavage once per day during 3 consecutive days. Efficiency of tamoxifen treatment was routinely performed by genotyping for the presence of the floxed allele.

For **paw oedema studies** *Pik3ca*^{RBD/-} and *Pik3ca*^{WT/-} mice⁶⁴ were divided into groups of four two weeks before edema induction. Before inducing the paw edema, the mice were anesthetized with 4% isoflurane. To induce the edema, mice received ipsilateral i.pl. injection (30 μl) of either zymosan (10 μg/μl, Sigma Aldrich) or PBS into the back-hind paw. Injection of 0.1mg/kg Buprenorphine (NOAH, Vetergesic) was given for pain prevention. Paw thickness was measured using a caliper every hour during the first 6 hours after injection and then at 8, 10 hours and twice per day afterwards. Buprenorphine was injected twice per day during the length of the experiment. Mice were kept, managed, and sacrificed in the NUCLEUS animal facility of the University of Salamanca according to current European (2007/526/CE) and Spanish (RD 1201/2005 and RD53/2013) legislation. All experiments were approved by the Bioethics Committee of the Cancer Research Center.

Isolation, culture and treatments of BMDM

Bone marrow cells from tibias and femurs of 12-14-week-old *Pik3ca*^{RBD/Lox} mice and *Pik3ca*^{WT/Lox} littermates were cultured with DMEM supplemented with 10% FBS, 100 units/ml penicillin, 100 μg/mL streptomycin, 2 mM L-Glutamine and 20 ng/mL M-CSF for 7 days. 4-hydroxytamoxifen (Sigma Aldrich) (100 nM) was added to culture media on day 3 to eliminate *Pik3ca*-Lox allele. The differentiated BMDM were then detached using cell dissociation buffer (C5914-100, Merck) and cultured in DMEM supplemented with 10% FBS, 100 μg/mL streptomycin, 2 mM L-Glutamine, 20

ng/mL M-CSF for unstimulated BMDMs. For macrophage polarization towards an inflammatory phenotype 20 ng/ml IFN γ (Peprotech) and 100 ng/ml LPS (Sigma Aldrich) was added to culture media.

Generation of chimeric animals

Chimeric mice exhibiting WT or RBD deficient leukocytes were generated by lethal irradiation with 5.5 Gy twice, 4 h apart of LysM GFP recipient animals (mice exhibiting endogenously GFP fluorescent monocytes and neutrophils) followed by an injection of bone marrow cells (1.5×10^6 cells/recipient i.v.) from C57BL/6 WT or RBD donor mice. Chimerism was then assessed 4 weeks later by flow cytometry from blood samples (reconstitution of $99.8 \pm 0.2\%$ and $99.8 \pm 0.1\%$ for WT and RBD deficient donor cells, respectively; n =5 mice per group).

Neutrophil depletion

Neutrophil depletion of chimeric mice was induced by intraperitoneal injection of anti-GR1 25 μ g/mouse/day for 3 days). Numbers of blood circulating monocytes and neutrophils were quantified by flow cytometry pre- and post-depletion. Neutrophils were found to be reduced by 99.5%, whilst this anti GR1-depleting protocol had no effect on blood monocyte proportion (n=5 mice/group).

Brightfield intravital confocal microscopy

Mesenteric inflammation was induced following intraperitoneal injection of mouse recombinant CCL2 (500ng/mouse in 500 μ L of PBS). Six hours later, anesthetized chimeric mice (150 mg/kg ketamine, 7.5 mg/kg xylazine, i.p.) were placed in supine position on a heating pad (37 °C) for maintenance of body temperature. The mesenteric vascular bed was exteriorized, placed on a purpose-built stage of an upright brightfield microscope (Zeiss Axioskop). Mesenteries were superfused with warmed (37 °C) Tyrode's solution (Sigma). After a 5-min equilibration period, analysis of leukocyte-endothelium interactions was made in at least 9 (and up to 16) randomly selected segments (100 μ m in length) of post-capillary venules (20–40 μ m in diameter) for each mouse. Leukocyte rolling was quantified by counting the number of rolling cells passing a fixed transversal line in the middle of the vessel segment for 5 min. Leukocyte adhesion (stationary position of the cell for 30 s or longer) was quantified along a 100 μ m vessel length and data were normalized as the number of cells per 500 μ m vessel segments. Leukocyte extravasation response was quantified within 50 μ m on either side of the 100 μ m vessel segment in the perivenular tissue; and data were normalized as the number of extravasated leukocytes per mm² of extravascular tissue. At the end of the analysis period, mice were humanely killed by cervical dislocation.

Immunofluorescence

BMDM were fixed using 4% paraformaldehyde, permeabilized with 0.1% Triton X-100 and blocked for 1 h with 3% BSA in PBS before incubation with the primary antibodies, used at a 1:100 dilution: Lamp1 (#553792, Bdbio), Deoxyribonuclease I-Alexa Fluor™ 488 Conjugate (#D12371, Invitrogen, 1:2000). To stain actin cytoskeleton, Alexa Fluor™ 647 Phalloidin (Invitrogen, 1:10 000) was directly added to the primary antibody mixture. Alexa Fluor 488- or Alexa Fluor 555-conjugated secondary antibodies (Invitrogen) were used to detect the indicated proteins at a 1:1000 dilution. Cells were counterstained with DAPI on the mounting solution (ProLong Gold Antifade Reagent with DAPI, Invitrogen). Images were taken using a Zeiss LSM510 confocal microscope or Leica DM6 B THUNDER Imager 3D Tissue.

Transwell migration assay

Transwell migration assays were carried out using the 6.5 mm Transwell® with 8.0 μ m Pore Polyester Membrane Insert (Corning). 9×10^4 MEFs from wild type mice were used as a chemo-attractant to encourage macrophage migration. 8×10^5 BMDM were seeded in the transwell.

Transwells were performed following the manufacturer instructions.

Random migration assay

For random migration assays, BMDMs were seeded in 24-well plates coated with matrigel (0.5mg/ml) and labelled using CellTracker™ Red CMTPX Dye 1 μ M (ThermoFisher) for 30 minutes. 24 hours later LPS + IFN- γ was added when necessary. Triplicates of each condition and genotype were prepared. Time-lapse imaging was carried out for 24 h. One image was taken every 10 min within the same well using a Nikon microscope driven by Metamorph (Molecular Devices, Chicago, IL, USA). A total of 80-100 cells per condition were tracked using the Fiji plugin Trackmate. Pre-processing was done using Mexican hat filter 3.0 radius to increase particle detection. Images were segmented using the fluorescence channel with the Laplacian of the Gaussian detector with a 30 μ m estimated particle diameter, a 10.0 threshold and median filter option selected. Segmented objects were linked from frame to frame with a Linear Assignment Problem (LAP) tracker with 45 μ m frame-to-frame linking distance and 2 frame gap closure. Criteria for track acceptance were track duration at least the 90% of the video. Tracks were visually inspected for completeness and accuracy of the tracking.

Secretome mass spectrometry

Samples for secretome analysis were prepared as previously described¹³⁰. In brief, 100ug of proteins were digested into peptides using trypsin and peptides were desalted using Oasis HLB extraction cartridges (Waters UK Ltd)) and eluted with 50% acetonitrile (ACN) in 0.1% Trifluoroacetic acid (TFA).

Dried peptides were dissolved in 0.1% TFA and analysed by nano ACQUITY liquid chromatography (Waters Corp., Milford, MA, USA) coupled on-line to a tandem LTQ Orbitrap XL, mass spectrometer (Thermo Fisher Scientific)¹³¹. Gradient elution was from 5% to 25% buffer B in 180 min at a flow rate 300nL/min with buffer A being used to balance the mobile phase (buffer A was 0.1% formic acid in water and B was 0.1% formic acid in ACN). The mass spectrometer was controlled by Xcalibur software and operated in the positive mode. The spray voltage was 1.95 kV and the capillary temperature was set to 200 °C. The LTQ Orbitrap XL was operated in data dependent mode with one survey MS scan followed by 5 MS/MS scans. Label-free quantitative proteomics analysis was performed using three independent biological samples per group. Additionally, each sample was analysed in technical duplicates. To ensure robust quantitative analysis, we utilized LTQ Orbitrap XL tandem mass spectrometry (MS/MS) to generate six distinct mass spectral profiles from each group.

MS raw files were converted into Mascot Generic Format using Mascot Distiller (version 2.3.0) and searched against the SwissProt database (release December 2015) restricted to human entries using the Mascot search daemon (version 2.3.1). Allowed mass windows were 10 ppm and 600 mmu for parent and fragment mass to charge values, respectively. Variable modifications included in searches were oxidation of methionine, pyro-glu (N-term) and phosphorylation of serine, threonine and tyrosine.

Spectral counting quantification method relies on the number of times peptides are identified by tandem mass spectrometry (with expectancy value <0.05) from a given protein. Spectral counts were obtained from Mascot result (DAT) files using a python script written in house in the Mascot Parser Toolkit environment (version 2.4.x).

Proteomic data analysis

The proteomic data obtained consisted of 6,844 peptides and 30 samples: Pik3ca^{WT/-} and Pik3ca^{RBD/-} BMDMs in steady state conditions (labelled as cell samples: WT/- and RBD/-), phagocytosing apoptotic cells (labelled as cell samples: WT/-Phag and RBD/-Phag), and the

apoptotic LKR10 cells alone (labelled as LKR). For each of these samples, proteomic experiments were performed with 6 replicates (3 biological replicates x 2 technical replicates), yielding a data set of 30 samples.

The first analytical step was to remove all peptides for which there was no information contained in the proteomic raw data matrix and the peptides for which 85% or more of the signal values were missing. All these peptides were specific of mouse proteins and in many cases were unique. The corresponding proteins were annotated and labelled together with each measured peptide. Next, low-quality samples were also removed, testing the overall signal per sample to identify if there were clear outliers with very low signal or with a very different signal distribution. Comparison of the overall signal distributions of the 30 samples (comparing boxplots) and identified 3 samples that were very different were obtained and discarded (WT/-_s3r2 (sample 3, replicate 2), RBD/-_s2r1 and LKRc_s1r1). These 3 samples showed a median signal in their distributions that deviated >20% from the median signal of the distributions of all other samples.

Differential expression analysis for each peptide of each protein were next performed. The algorithm used to carry out this analysis was *limmaVoom* within EdgeR R package⁶⁵,⁶⁶. Prior to this analysis, a Bartlett test was performed to see the homogeneity of variances, verifying that for this data we cannot consider equality of variances and this factor was included in the differential expression algorithm. With this algorithm, normalization factors to use *a-posteriori* were calculated and, transformation and calculation of the variance weights was performed. The model to fit before using *Voom* as specified since it uses the variances of the model's residuals (observed - fitted). Finally, an estimation of the contrast for each feature tested (i.e. each peptide) was carried out using the *Empirical Bayes* approach in *limma* as previously described⁶⁷. Peptides were ordered by the p.value of the *limma* test considering significant peptides changed only with a p.value below 0.05 and with a log2(Fold-Change) > |2|. All these analyses were performed using the statistical computing language R and packages or libraries obtained from R-cran (cran.r-project.org) or Bioconductor (www.bioconductor.org).

Cytoscape software (v3.9)⁶⁸ including GeneMania app⁶⁹ was then used to generate and visualize protein-protein networks of the significantly altered proteins selected in secretome analysis of unstimulated and phagocytosing BMDMs. This tool provides information on protein-protein associations based in co-expression studies and also based in physical interaction studies.

Western blot analysis

Immunoblot was performed per a general western-blot protocol (Abcam). Total protein was extracted using Cell Lysis Buffer (Cell Signaling Technology) supplemented with cOmplete mini protease inhibitor cocktail (Roche), 50 mM sodium fluoride and 1 mM of PMSF. Protein was quantified using Bradford Method (Bio-Rad). 20 µg of protein was separated by SDS-PAGE and transferred to 0.2 µm pore-size PVDF membranes (Sigma-Aldrich). Blots were probed using the following antibodies, at a concentration 1:1000 unless otherwise stated: phospho-LIMK1 (Thr508)/LIMK2 (Thr505) (#3841, Cell signalling), cofilin (sc-376476, Santa Cruz Biotechnology), phospho-cofilin (Ser3) (#3311, Cell Signalling), cathepsin B (12216-1-AP, Proteintech), cathepsin D (21327-1-AP, Proteintech), α-tubulin (ab15246, Abcam; concentration 1:5000). Horseradish peroxidase-conjugated secondary antibodies (Amersham) were used (1:5000) and detected using an enhanced chemiluminescent substrate (Amersham). Signal was detected using an iBright 1500 System (Invitrogen).

Flow cytometry analysis

Single-cell suspensions from cultured cell, spleen or blood monocytes were generated from mice, washed twice in staining buffer and incubated with 1:100 Fc-block (BD Biosciences, #553142) diluted in FACS buffer. Cells were subjected to surface antibody staining with labelled antibodies

diluted in staining buffer for 30 min at 4 °C: CD3-PE-Cy7 (#100328, Biolegend), CD4-BV605 (#100548, Biolegend), CD8-APC (#100712, Biolegend), CD19-PerCP-Cy5.5 (#115534, Biolegend), CD45-BV785 (#103149, Biolegend), Ly6C-PerCP-Cy5.5 (#128012, Biolegend), Ly6C-E450 (#48-5932-82, eBioscience), Ly6G-AF700 (#56-5931-82, eBioscience), CD11b-BV650 (#101239, Biolegend), F4/80-PE-Cy7 (#123114, Biolegend), F4/80-PE (#123110, Biolegend), CCR2 (CD192)-PE-Vio @ 770 (#130-108-724, Miltenyi Biotec). After incubation, cells were washed in staining buffer and analysed immediately. For all staining, isotype controls were used.

Samples were acquired on a BD LSR FORTESA FACS or FACS Aria III machine that uses FACS DIVA software (BD Biosciences). Compensation was performed using 1 drop of Ultracomp ebeads (eBioscience) in 300µl of FACS buffer. 1µl of each antibody used in the pool was mixed with 100µl of compensation beads solution and acquired. Analysis was performed with FlowJo software (FlowJo V10.4). Once the different pools were compensated samples were acquired.

Phagocytosis assay

For phagocytosis assay, BMDMs in suspension in DMEM without FBS were labelled with 1:200 red cell tracker (Molecular Probes) for 30 minutes at 37°C. Cells were then centrifuged for 5 minutes at 300g at 4°C, supernatant was removed and labelled BMDMs were washed twice with 5ml of PBS and plated overnight in complete DMEM containing 20 ng/ml M-CSF. On the following day, 100 ng/ml LPS (Sigma Aldrich) was added to BMDMs overnight.

LKR10 cells (murine lung cancer cell line) were stained with red cell tracker (Molecular Probes) as previously described for macrophages. Stained cells were plated in complete DMEM medium and, after 12 hours, 50 µM Cisplatin (MCE MedChemExpress) was added to the media and left overnight. BMDMs were then incubated with apoptotic cancer cells at a 1:2 ratio and cultured at 37 °C for different time points in DMEM supplemented with 10% FBS.

Flow cytometry data were acquired using a BD LSR FORTESA FACS instrument with FACS DIVA software (BD Biosciences) and analysed using FlowJo V10.4 software. A minimum of 2×10^5 events were acquired and analysed. Data analysis and interpretation was done using FlowJo software (FlowJo V10.4).

Image analysis

Confocal images were post-processed and analysed using Fiji distribution of ImageJ version 1.53q. Cell shape descriptors such as “aspect ratio” (AR), “circularity” (C) and “cell area” were measured using Fiji. Specifically, aspect ratio is calculated as $(\text{major axis} \times \text{minor axis} - 1)$ therefore representing solely the degree of elongation, whereas circularity is calculated as $[4\pi \times (\text{area} \times \text{perimeter} - 2)]$, thus representing the degree of similarity to a circumference with a value ranging from 0 to 1 (perfect circle).

Histology

Tissue was fixed using 4% formaldehyde for 48h, dehydrated and paraffin-embedded. Sections (3 µm) were cut and stained using hematoxylin-eosin. For immunodetection, citrate pH 6 buffer was used for antigen retrieval. Staining was used using the following primary antibodies: CD68 (ab125212, Abcam, 1:200), cathepsin B (12216-1-AP, Proteintech, 1:100), cathepsin D (21327-1-AP, Proteintech, 1:400). Dako EnVision+ System HRP labelled Polymer secondary antibodies (Dako) were used, and DAB+ Substrate Chromogen System (Dako) was used for color development.

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Declarations

Authors Disclosures

The authors declare no conflict of interest.

Authors Contributions

Conception and design: E. Castellano. **Development of methodology:** A. Rosell, M. Alcón Pérez, M.B. Voisin, J. de Paz, V. Rajeeve, C. Cuesta, C. Morillo, O. Swinyard, E. Gabandé-Rodriguez.

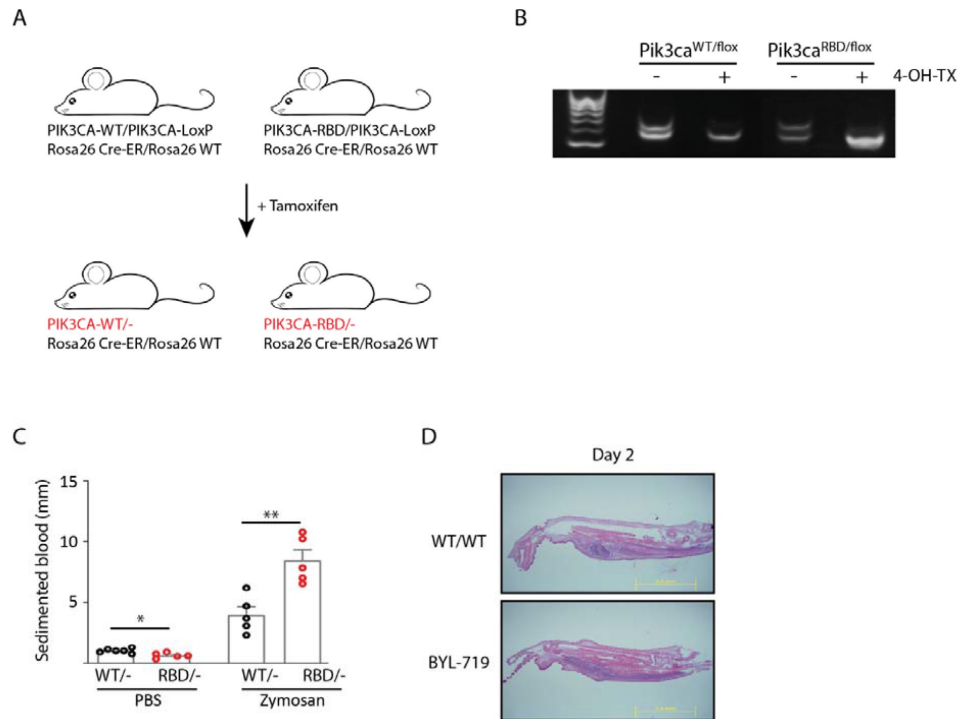
Acquisition of data (provided animals, acquired and managed patients, provided facilities, etc.): A. Rosell, M. Alcón Pérez, A. Krywoska, M.B. Voisin, J. Downward, P. Cutillas, E. Castellano.

Analysis and interpretation of data (e.g., statistical analysis, biostatistics, computational analysis): A. Rosell, M. Alcón Pérez, A. Krywoska, M.B. Voisin, V. Rajeeve, A. Berral-Gonzalez, J. De Las Rivas, P. Cutillas, E. Castellano. **Writing, review, and/or revision of the manuscript:** A. Rosell, M. Alcón Pérez, M.B. Voisin, J. De Las Rivas, P. Cutillas, E. Castellano.

Supplementary Figures

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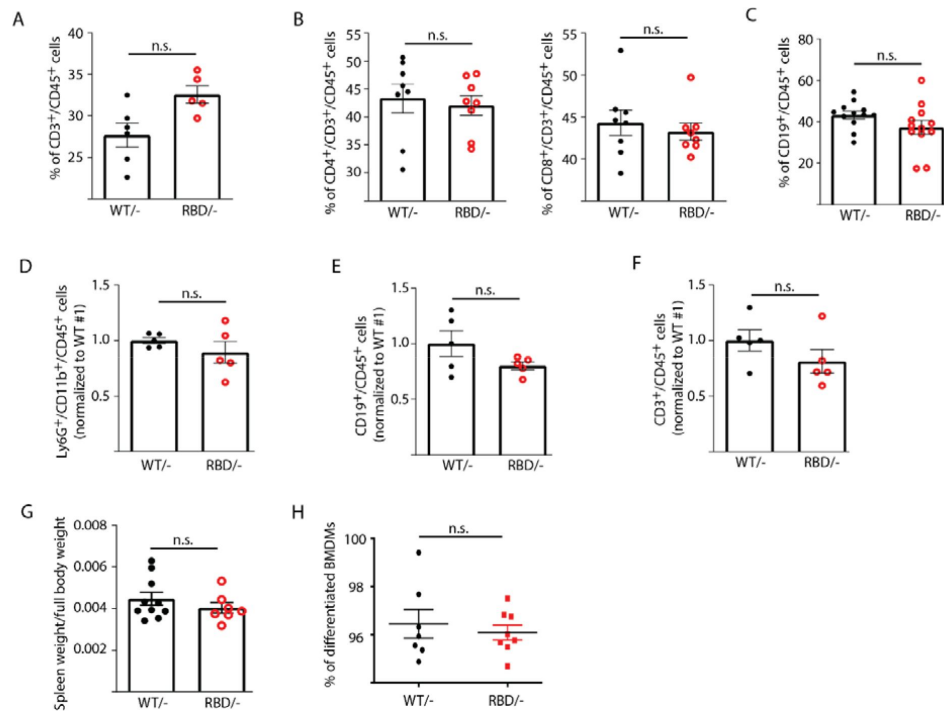
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Supplementary Figure S1.

Pik3ca^{RBD/-} mice have more extended inflamed tissue after zymosan injection.

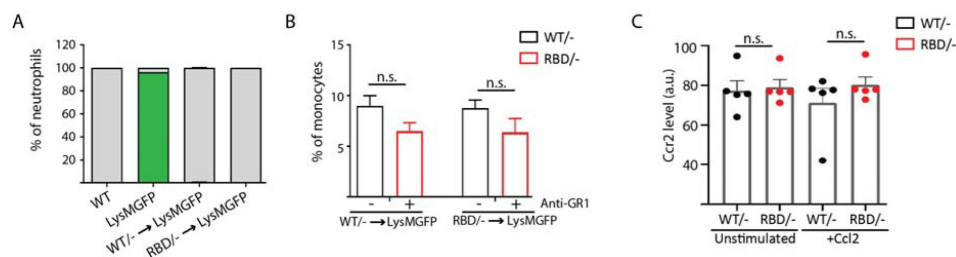
(A) Schematic representation of the mouse model used in which the interaction of p110 α with RAS was disrupted by the introduction of two-point mutations, T208D and K227A, into the endogenous *Pik3ca* gene (*Pik3ca*^{RBD}). Wild-type (*Pik3ca*^{WT}) and *Pik3ca*^{RBD} mice were bred with mice containing a floxed *Pik3ca* allele and a mouse carrying a conditional Cre recombinase (Cre-ERT2) allele targeted to the ubiquitously expressed *Rosa26* locus; **(B)** Representative PCR showing efficiency of floxed allele removal after treatment with 4-hydroxytamoxifen for 48 hours; **(C)** Blood from the same mice was obtained by cardiac puncture prior culling and blood sedimentation was measured. The clear layer formed at the top was used for data quantification. Black dots represent data from *Pik3ca*^{WT/-} mice and red dots represent data from *Pik3ca*^{RBD/-} mice. Each dot represents an individual mouse. Error bars indicate mean $\pm$ SEM. Significance using Student's t test: * p < 0.05; ** p < 0.01. *Pik3ca*^{WT/-} PBS n = 6; *Pik3ca*^{RBD/-} PBS n = 5; *Pik3ca*^{WT/-} Zymosan n = 4; *Pik3ca*^{RBD/-} Zymosan n = 4; **(D)** Representative images showing zymosan-induced inflammatory area in *Pik3ca*^{WT/WT} mice and *Pik3ca*^{WT/WT} mice treated with BYL-719.



Supplementary Figure S2.

Disruption of RAS-p110α interaction does not affect the number of B-cells or T-cells in blood or spleen.

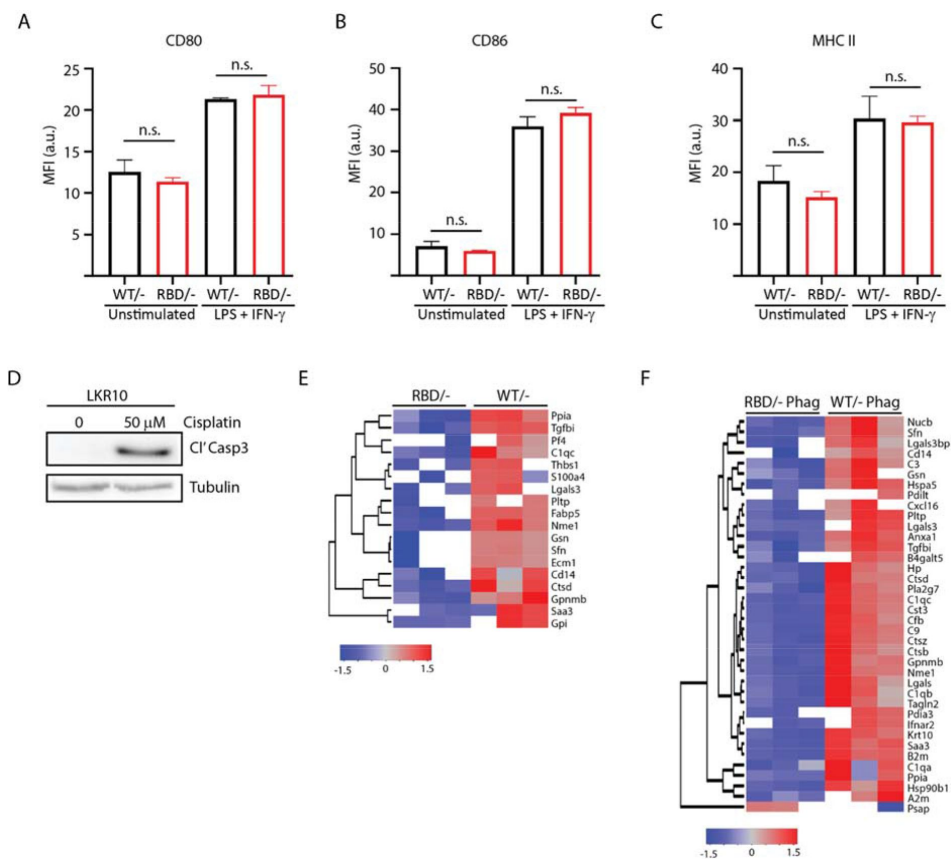
Twelve-week-old mice were treated with tamoxifen and after four weeks, flow cytometry analysis was performed to determine abundance of different immune populations. **(A)** CD3 positive cells in circulating blood; **(B)** CD4⁺ and CD8⁺ T-cells in circulating blood; **(C)** B-cells in circulating blood; **(D)** Granulocytes in spleen; **(E)** B-cells in spleen; **(F)** T-cells in spleen; **(G)** Spleen weight compared to full body; **(H)** Graph depicting % of differentiated macrophages obtained in culture from bone marrow precursors in the presence of m-CSF. Black dots represent data from *Pik3ca*^{WT/-} mice and red dots represent data from *Pik3ca*^{RBD/-} mice. Each dot represents an individual mouse. Error bars indicate mean ± SEM. Significance using Student's t test: n.s. non-significant.



Supplementary Figure S3.

Pik3ca^{RBD/-} monocytes do not extravasate through the endothelium.

Graph showing levels of neutrophil reconstitution in chimera's bone marrow. Blood from chimera mice was collected and GFP in granulocytes was analysed by flow cytometry to determine engraftment percentage. **(B)** Graph quantifying the number of monocytes in the blood of chimera mice treated with anti-GR1 (25µg/mouse/day) for 3 days. **(C)** Graph showing levels of expression of CCR2 in *Pik3ca*^{WT/-} and *Pik3ca*^{RBD/-} BMDMs detected by flow cytometry. Error bars indicate mean ± SEM. Significance using Mann-Whitney test: n.s. no-significant



Supplementary Figure S4.

Disruption of RAS-p110 α interaction alters the secretome of BMDMs.

Pik3ca^{RBD/-} and *Pik3ca*^{RBD/-} BMDMs were stimulated with LPS+IFN- γ and expression of (A) CD80; (B) CD86; and (C) MHCII were analysed by flow cytometry (*Pik3ca*^{RBD/-} n = 5; *Pik3ca*^{WT/-} n = 3). Error bars indicate mean $\pm$ SEM. Significance using Mann-Whitney test: n.s. no-significant. (D) WB showing induction of apoptosis in LKR10 cell line treated with cisplatin for 16 hours; (E) Dendrogram showing hierarchical clustering of peptides found in the secretome analysis of *Pik3ca*^{RBD/-} and *Pik3ca*^{RBD/-} BMDMs in steady state conditions; (F) Dendrogram showing hierarchical clustering of peptides found in the secretome analysis of *Pik3ca*^{RBD/-} and *Pik3ca*^{RBD/-} BMDMs phagocytosing apoptotic cells.

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Article and author information

Esther Castellano

Instituto de Biología Molecular y Celular del Cáncer

For correspondence: ecastellano@usal.es

ORCID iD: [0000-0002-8449-4081](https://orcid.org/0000-0002-8449-4081)

Alejandro Rosell

Instituto de Biología Molecular y Celular del Cáncer

Agata Krygowska

Queen Mary University of London

Marta Alcón Pérez Alcón

Instituto de Biología Molecular y Celular del Cáncer

Mathieu-Benoit Voisin

Queen Mary University of London-William Harvey Research Institute

Juan de Paz

Instituto de Biología Molecular y Celular del Cáncer

Vinothini Rajeeve

Barts Cancer Institute, QMUL

ORCID iD: [0000-0002-6361-4291](https://orcid.org/0000-0002-6361-4291)

Alberto Berral-González

Instituto de Biología Molecular y Celular del Cáncer

Cristina Cuesta

Instituto de Biología Molecular y Celular del Cáncer

Ottillie Swinyard

Queen Mary University of London

Enrique Gabandé-Rodríguez

Queen Mary University of London

ORCID iD: [0000-0002-9715-8714](https://orcid.org/0000-0002-9715-8714)

Julian Downward

The Francis Crick Institute

ORCID iD: [0000-0002-2331-4729](https://orcid.org/0000-0002-2331-4729)

Javier De Las Rivas

Instituto de Biología Molecular y Celular del Cáncer

Pedro Cutillas

Barts Cancer Institute, QMUL

ORCID iD: [0000-0002-3426-2274](https://orcid.org/0000-0002-3426-2274)

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Editors

Reviewing Editor

Tadatsugu Taniguchi

University of Tokyo, Tokyo, Japan

Senior Editor

Tadatsugu Taniguchi

University of Tokyo, Tokyo, Japan

Reviewer #1 (Public Review):

In this study, Alejandro Rosell et al. uncovers the immunoregulation functions of RAS-p110 α pathway in macrophages, including the extravasation of monocytes from the bloodstream and subsequent lysosomal digestion. Disrupting RAS-p110 α pathway by mouse genetic tools or by pharmacological intervention, hampers the inflammatory response, leading to delayed resolution and more severe acute inflammatory reactions. The authors proposed that activating p110 α using small molecules could be a promising approach for treating chronic inflammation. This study provides insights into the roles and mechanisms of p110 α on

macrophage function and the inflammatory response, while some conclusions are still questionable because of several issues described below.

(1) Fig. 1B showed that disruption of RAS-p110 α causes the decrease in the activation of NF- κ B, which is a crucial transcription factor that regulates the expression of proinflammatory genes. However, the authors observed that disruption of RAS-p110 α interaction results in an exacerbated inflammatory state *in vivo*, in both localized paw inflammation and systemic inflammatory mediator levels. Also, the authors introduced that "this disruption leads to a change in macrophage polarization, favouring a more proinflammatory M1 state" in introduction according to reference 12. The conclusions drew from the signaling and the models seemed contradictory and puzzling. Besides, it is not clear why the protein level of p65 was decreased at 10' and 30'. Was it attributed to the degradation of p65 or experimental variation?

(2) In Fig 3, the authors used bone-marrow derived macrophages (BMDMs) instead of isolated monocytes to evaluate the ability of monocyte transendothelial migration, which is not sufficiently convincing. In Fig. 3B, the authors evaluated the migration in Pik3caWT/- BMDMs, and Pik3caWT/WT BMDMs treated with BYL-719'. Given that the dose effect of gene expression, the best control is Pik3caWT/- BMDMs treated with BYL-719.

(3) In Fig. 4E-4G, the authors observed that elevated levels of serine 3 phosphorylated Cofilin in Pik3caRBD/- BMDMs both in unstimulated and in proinflammatory conditions, and phosphorylation of Cofilin at Ser3 increase actin stabilization, it is not clear why disruption of RAS-p110 α binding caused a decrease in the F-actin pool in unstimulated BMDMs?

<https://doi.org/10.7554/eLife.94590.1.sa1>

Reviewer #2 (Public Review):

Summary:

Cell intrinsic signaling pathways controlling the function of macrophages in inflammatory processes, including in response to infection, injury or in the resolution of inflammation are incompletely understood. In this study, Rosell et al. investigate the contribution of RAS-p110 α signaling to macrophage activity. p110 α is a ubiquitously expressed catalytic subunit of PI3K with previously described roles in multiple biological processes including in epithelial cell growth and survival, and carcinogenesis. While previous studies have already suggested a role for RAS-p110 α signaling in macrophages function, the cell intrinsic impact of disrupting the interaction between RAS and p110 α in this central myeloid cell subset is not known.

Strengths:

Exploiting a sound previously described genetically mouse model that allows tamoxifen-inducible disruption of the RAS-p110 α pathway and using different readouts of macrophage activity *in vitro* and *in vivo*, the authors provide data consistent with their conclusion that alteration in RAS-p110 α signaling impairs the function of macrophages in a cell intrinsic manner. The study is well designed, clearly written with overall high-quality figures.

Weaknesses:

My main concern is that for many of the readouts, the difference between wild-type and mutant macrophages *in vitro* or between wild-type and Pik3caRBD mice *in vivo* is rather modest, even if statistically significant (e.g. Figure 1A, 1C, 2A, 2F, 3B, 4B, 4C). In other cases, such as for the analysis of the H&E images (Figure 1D-E, S1E), the images are not quantified, and it is hard to appreciate what the phenotype in samples from Pik3caRBD mice is or whether this is consistently observed across different animals. Also, the authors claim there

is a 'notable decrease' in Akt activation but 'no discernible change' in ERK activation based on the western blot data presented in Figure 1A. I do not think the data shown supports this conclusion.

To further substantiate the extent of macrophage function alteration upon disruption of RAS-p110 α signaling, the manuscript would benefit from testing macrophage activity in vitro and in vivo across other key macrophage activities such as bacteria phagocytosis, cytokine/chemokine production in response to titrating amounts of different PAMPs, inflammasome function, etc. This would be generally important overall but also useful to determine whether the defects in monocyte motility or macrophage lysosomal function are selectively controlled downstream of RAS-p110 α signaling.

Furthermore, given the key role of other myeloid cells besides macrophages in inflammation and immunity it remains unclear whether the phenotype observed in vivo can be attributed to impaired macrophage function. Is the function of neutrophils, dendritic cells or other key innate immune cells not affected?

Compelling proof of concept data that targeting RAS-p110 α signalling constitutes indeed a putative approach for modulation of chronic inflammation is lacking. Addressing this further would increase the conceptual advance of the manuscript and provide extra support to the authors' suggestion that p110 α inhibition or activation constitute promising approaches to manage inflammation.

Finally, the analysis by FACS should also include information about the total number of cells, not just the percentage, which is affected by the relative change in other populations. On this point, Figure S2B shows a substantial, albeit not significant (with less number of mice analysed), increase in the percentage of CD3⁺ cells. Is there an increase in the absolute number of T cells or does this apparent relative increase reflect a reduction in myeloid cells?

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